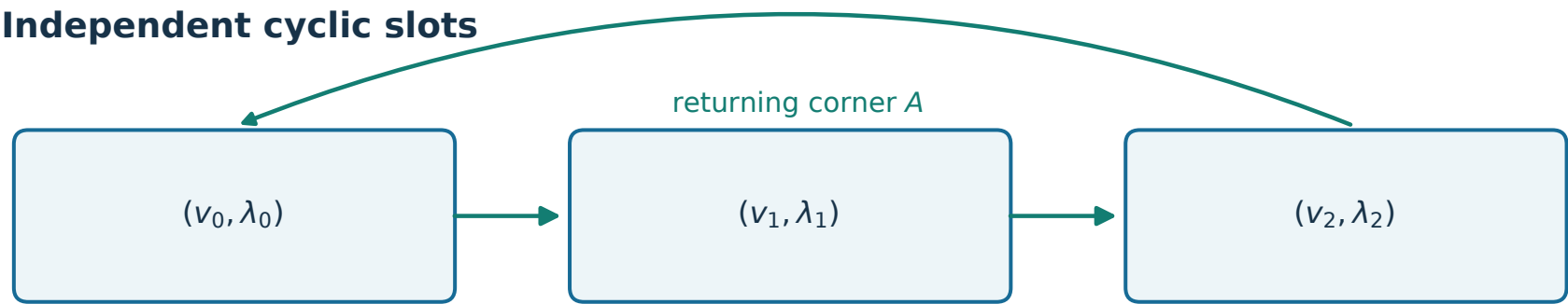


A returning zero is not an absent trace

Actual odd receiver at $d = 3$: $A = -F_p^{-1}$, original blocks of dimension $q = (k + 1)^2$.

Independent cyclic slots



$$\text{Tr}(C_3^n) = \tau \quad (3 \nmid n)$$

$$\text{Tr}(C_3^{3m}) = (3\text{tr}(A^m), 1_L)$$

If the returning amplitude is zero, this is e, not τ .

The exact synchronization retract



$$rj = I, \quad jr : (\lambda_j)_j \mapsto (\bigvee_j \lambda_j)_j$$

Global synchronization gives trace e also when $3 \nmid n$.

The cycle comparison retains $\Theta_3 - \Theta_1$, with ghost $3\mathbf{1}_{3 \mid n} - 1$.

Proof: FLW7-17, FLW23-31. Every original metric entry and mass factor is retained.
 Human source: Alain Connes and Caterina Consani, arXiv:2004.08879v1, lines 454-600.